CST538 - Advanced Software Engineering

CST 538 (Advanced Software Engineering) Syllabus

Course Information

Course Details

- **Course Title:** Advanced Software Engineering
- **Course Number:** CST 538
- **Meeting Location:** Physical
- **Units:** 4

Course Description

Students work in teams to build modern, concurrent, systems following industry best practices such as: CI/CD pipelines, test driven development, containerization, and version control. Students will develop a pitch for a project and create a fully functioning demonstration including documentation and a testing suite.

Materials

- Course textbook: "Clean Architecture: A Craftsman's Guide to Software Structure and Design" by Robert C. Martin

Technology Requirements

Students must have access to: - A computer (not "chromebook" or similar) with reliable internet connection - Access to Canvas learning management system

Course Objectives

At the end of this class, you should be able to:

1. Implement CI/CD pipelines and automated testing workflows in team environments
2. Apply test-driven development practices to build robust software systems
3. Utilize containerization technologies for deployment and development
4. Collaborate effectively using modern version control and project management tools

Grading

There are three groups of assignments, briefly described in the table below. **Getting below a 40% in any of these three groups will result in failing the class.**

Assignment Kind	Value
Projects	50%
Exams	20%
Participation	30%

Projects

Projects will take the form of team-based software development with CI/CD pipelines, containerized application deployment, and collaborative coding using modern version control practices. Students will pitch, design, and implement fully functioning systems with comprehensive testing suites and documentation.

Exams

Exams will cover material from class and from any assigned reading, as well as draw from knowledge gained during projects.

Participation

Class is expected to be highly interactive, with students actively answering and asking questions, as well as participating in discussion groups. Therefore, attendance is required, as well as participation in all in-class activities. These activities may take the form of in-class discussions, brainstorming sessions, stand-ups, as well as making active progress on projects during class time.

Grade Assignment

Grades as assigned based on the standard ranges, which are outlined below.

Grade Range	Letter Grade
$[97, \infty)$	A+
$[93, 97)$	A
$[90, 93)$	A-
$[87, 90)$	B+
$[83, 87)$	B
$[80, 83)$	B-
$[77, 80)$	C+
$[73, 77)$	C
$[70, 73)$	C-
$[60, 70)$	D
$[0, 60)$	F

Syllabus Change Policy

This syllabus is subject to change at the instructor's discretion to enhance learning or accommodate unforeseen circumstances. Students will be notified of any substantive changes (such as changes to due dates, point values of assignments, or major policy modifications) via Canvas announcement and/or email at least one week in advance when possible.

The process for syllabus changes: 1. Instructor identifies need for change 2. Revised syllabus section is prepared 3. Students are notified via Canvas and email 4. Change takes effect after notification period

Students are responsible for staying current with any announced changes to the syllabus.